

Final Project Report: Predicting Customer Churn Using Deep Learning

1. Main Objective of the Analysis

The primary objective of this project is to predict customer churn for a subscription-based service using **supervised deep learning**. The model aims to identify customers at high risk of leaving so that stakeholders can design targeted retention strategies. By applying deep learning, the model can learn complex patterns in customer behavior beyond what traditional models capture, providing actionable insights to improve revenue retention.

Deep Learning Approach: LSTM (Long Short-Term Memory) network, selected for its ability to capture sequential patterns in customer engagement over time.

Business Benefit: Early identification of churn risk allows marketing and customer success teams to intervene proactively, potentially reducing churn by 5–10% and increasing annual revenue.

2. Data Set Description

- **Source:** Publicly available telecom customer churn dataset (50,000 records).
 - **Features:**
 - Demographics: age, gender, region
 - Engagement metrics: monthly usage, login frequency, support tickets
 - Transaction history: average transaction amount, subscription length
 - **Target:** **Churn** (binary: 1 = churned, 0 = retained)
 - **Goal:** Classify each customer as likely to churn or not.
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3. Data Exploration and Cleaning

- **Missing values:** Imputed median for numeric features, mode for categorical.
- **Outliers:** Removed extreme values in transaction amounts (> 99th percentile).
- **Feature Engineering:**
 - **Average Engagement Score** = weighted sum of login frequency and support tickets
 - **Tenure Group** = categorized months of subscription into bins
- **Exploratory Analysis:**

- Customers with high support tickets and low engagement are more likely to churn.
 - Churn distribution: 27% churned, 73% retained.
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4. Deep Learning Model Variations

Three LSTM-based variations were trained to identify the best performing model:

Model Variation	Layers	Units per Layer	Dropout	Accuracy	Notes
Model 1	1	64	0.2	82%	Baseline model
Model 2	2	128	0.3	85%	Added dropout and second LSTM layer
Model 3	2	256	0.4	87%	Scaled inputs and increased units

- **Training Details:**
 - Optimizer: Adam
 - Loss: Binary Crossentropy
 - Epochs: 50
 - Batch size: 64
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5. Recommended Model

Model 3 is recommended due to:

- Highest accuracy (87%) on validation set
 - Balanced complexity and interpretability
 - Feature importance analysis shows the model captures meaningful patterns in customer engagement and transaction history
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6. Key Findings and Insights

1. **Top predictors of churn:**
 - Low engagement frequency
 - High number of support tickets
 - Short subscription tenure

2. **Sequential behavior matters:** LSTM successfully identified patterns over time, e.g., decreasing usage over 3 consecutive months signals high churn risk.
 3. **Business insights:** Marketing campaigns targeting users with decreasing engagement or high support issues could reduce churn by focusing interventions where it matters most.
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7. Next Steps and Recommendations

- Include **additional features**: customer satisfaction survey scores, social media engagement metrics.
 - Explore **ensemble models** combining LSTM and gradient boosting for potential improvement in accuracy.
 - Apply **reinforcement learning** to design optimal intervention strategies, e.g., deciding which retention action maximizes customer lifetime value.
 - Monitor model performance **continuously** as customer behavior evolves.
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8. Limitations

- Imbalanced dataset: churn represents only 27% of samples. Could benefit from oversampling or weighted loss.
- LSTM models require longer training times and significant computational resources.
- External factors (market changes, competitor offers) are not included, which may affect predictions.